

FINAL PROJECT REPORT

Power BI Data Analytics Project

Smartphone Usage, Digital Addiction & Well-Being Analytics

*An interactive Power BI report analysing the screen-time, lifestyle,
stress and demographic drivers of smartphone addiction across 7,500 users*

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1. Introduction

1.1 Project Overview

Smartphones have become the primary interface through which people work, study, socialise and relax. The same device that improves productivity also fragments attention through constant notifications, social-media scrolling and gaming. This project — Smartphone Usage, Digital Addiction & Well-Being Analytics — uses Microsoft Power BI to convert a raw usage dataset of 7,500 users into an interactive, multi-page analytical solution.

The dataset captures, for every user, demographic attributes (age, gender), behavioural attributes (daily screen time, weekend screen time, social-media hours, gaming hours, app opens per day, notifications per day, work/study hours) and well-being attributes (sleep hours, stress level, academic/work impact, addiction level and an addicted flag). The analysis reveals that 5,308 of the 7,500 users — 70.77% — are classified as addicted, and the report is designed to explain which habits sit behind that number.

1.2 Objectives

- Consolidate the smartphone_usage dataset into a single, clean and reliable Power BI data model.
- Quantify the scale of smartphone addiction in the user base and profile it by gender, age group and stress level.
- Identify which behavioural factors — weekend screen time, social-media hours, notifications or app opens — correspond most strongly with the addicted classification.
- Examine the relationship between screen-time behaviour and well-being outcomes such as sleep hours and academic/work impact.
- Deliver an interactive, slicer-driven, multi-page Power BI report that allows stakeholders to explore the findings themselves rather than reading a static summary.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The organization needs to understand why 5,308 of its 7,500 users (70.77%) are classified as addicted to smartphone usage, and which factors — screen-time behaviour, lifestyle habits, stress, or demographic characteristics — most strongly correspond with that classification. Without this understanding, product and research teams cannot prioritize well-being feature investment or design targeted digital-health interventions.

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a product / research team	understand why 5,308 of 7,500 users (70.77%) are classified as addicted	the overall addiction classification does not explain the key contributing factors	the available user data includes different behavioural, lifestyle, stress and demographic characteristics	uncertain about where to focus analysis and well-being efforts
PS-2	a product / research team	identify which factors most strongly correspond with smartphone addiction	it is difficult to prioritize well-being features and targeted interventions without clear evidence	screen-time behaviour, lifestyle habits, stress and demographics may have different relationships with addiction	concerned that digital-health interventions may not be targeted effectively

2.2 Project Proposal (Proposed Solution)

Section	Description
Objective	Design and deploy an interactive multi-page Power BI report that consolidates the smartphone usage data into a single unified workspace, enabling stakeholders to drill down into behavioural and well-being metrics and extract actionable insights.

Section	Description
Scope	Data ingestion via Power Query, data modelling within the Power BI Data Model layer, creation of calculated columns and DAX measures, and user-interface layouts implementing the Fluent2 theme across 5 analytical pages.
Approach	An Agile BI workflow: ETL and data cleansing in Power Query, deriving grouping columns (Age Group, Addicted Status, High Screen Time User), writing optimized DAX measures, and building slicer-driven canvases with cross-filtering.
Key Features	Interactive cross-filtering, decomposition-tree drill-down, Key Influencers AI visual, KPI card summary layer, and gender / age-group / addiction-level / stress-level slicers on every analytical page.
Impact	Replaces static spreadsheets with self-service exploration, shortens the reporting cycle and gives the well-being team direct evidence for prioritizing interventions.
Resources	Intel Core i7 / AMD Ryzen 7 workstation with 16 GB RAM and 512 GB NVMe SSD; Microsoft Power BI Desktop (latest); MySQL for exploratory queries; DAX and Power Query M as expression languages.

2.3 Initial Project Planning

Work was organised into four sprints comprising 11 user stories and 22 story points.

Sprint	Epic	USN	User Story / Task	Pts	Priority
Sprint-1	Setup	USN-1	Understand the problem — why 5,308 of 7,500 users are addicted (70.77%)	2	High
Sprint-1	Data Import	USN-2	Load 7,500 user records into Power BI with age, gender, screen time fields	2	High
Sprint-1	Data Clean	USN-3	Clean data and create Age Group (Under 20, 20–29, 30–39)	2	Medium
Sprint-2	KPI Cards	USN-4	Build 11 KPI cards — 7,500 total, 5,308 addicted, 26.57 age, 7.50 daily, 9.24 weekend, 3.27 social, 6.74 sleep, 134.26 notifications, 97.83 opens	3	High
Sprint-2	Report 1	USN-5	Build the KPI summary page (page 1 of 5)	1	High
Sprint-3	Charts	USN-6	Build the visualization page with 6 charts — gender, addiction level, donut, scatter, age group, gaming vs social	3	High
Sprint-3	Dashboard	USN-7	Build the dashboard — 4 cards and 6 charts including weekend screen time by addicted status and age-group split	3	High
Sprint-3	Filters	USN-8	Add slicers for gender, age group, addiction level and stress level	1	Medium
Sprint-4	Problem Statement	USN-9	Finalise PS-1 and PS-2 around the 20–29 cohort, 7.50 hrs daily and 9.24 hrs weekend usage	2	High
Sprint-4	Insights	USN-10	Identify strongest factors — weekend screen-time gap, then social media, then notifications	2	High
Sprint-4	GitHub README	USN-11	Prepare the README and screenshots for the repository	1	Medium

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The purpose of the collection plan is to identify the key raw data source, maintain data integrity and ensure that findings are valid. Data is retrieved directly from the shared course project folder containing the source usage logs and user well-being records.

Source Name	Description	Format	Size	Access
smartphone_usage dataset	Records for 7,500 users covering daily screen time, weekend screen time, social-media hours, gaming hours, app opens, daily notifications, sleep hours, stress level, academic/work impact, addiction level and addicted label.	CSV / Excel	~1.8 MB	Shared link access

3.2 Data Quality Report

Data Quality Issue	Severity	Resolution Plan
Raw age values were individual numbers, with no age brackets available for charting.	Moderate	Added a calculated column Age Group grouping ages into Under 20, 20–29 and 30–39.
The addiction column held raw codes instead of readable labels for report visuals.	High	Added a calculated column Addicted Status displaying "Addicted" / "Not Addicted".
No column existed to separate heavy phone users from normal users.	High	Added a calculated column High Screen Time User labelling each row "High Usage" or "Normal Usage".
No fixed total-user column existed to display counts on cards.	Moderate	Added a calculated column User Count to show the total of 7,500 users in cards and charts.
Academic and work impact values were mixed, making it hard to filter affected users.	Moderate	Added a conditional column tagging impact as High, Medium or Low Impact.
Stress level values were stored as plain numbers without categories.	Low	Added a column grouping stress into Low, Medium and High Stress.
Sleep duration hours were ungrouped, hiding who was getting enough rest.	Moderate	Added a custom column labelling Low Sleep (under 6 hours) or Normal Sleep (6+ hours).

3.3 Data Exploration and Preprocessing

Stage	Action Performed
Data Overview	Profiled all 7,500 rows and 15+ columns in Power Query, reviewing column distribution, column quality and column profile to confirm ranges for screen time (≈3–12 hrs), sleep (≈4.5–9 hrs) and age (mean 26.57).
Data Cleaning	Checked for NULLs in age, daily_screen_time_hours and addiction_level using SQL and Power Query; removed duplicate user records and blank rows so that every user_id appears exactly once.
Data Transformation	Applied Power Query steps to rename columns to friendly names, trim and clean text categories, and derive the grouping columns listed in the Data Quality Report.
Data Type Conversion	Re-cast usage-duration fields to decimal number, counts (app opens, notifications) to whole number, and categorical fields (gender, stress_level, addiction_level, academic_work_impact) to fixed text.
Data Modelling	Built the model around the smartphone_usage fact table with derived grouping columns acting as dimensions, so slicers on Age Group, gender, addiction level and stress level cross-filter every visual on the page.
Save Processed Data	Loaded the cleaned tables into the Power BI in-memory engine and saved the .pbix so that all five pages render from the same validated model.

4. Data Visualization

4.1 Framing Business Questions

No.	Business Question	Visualization Used
1	How does average daily screen time differ across genders?	Column chart — Average Daily Screen Time by Gender
2	What is the distribution of users across device addiction levels?	Bar chart — Users by Addiction Level
3	What proportion of users is addicted versus non-addicted?	Donut chart — Addicted vs Non-Addicted Users
4	Which age group shows the highest average daily screen time?	Column chart — Avg Daily Screen Time by Age Group
5	Is there a visible relationship between screen time and sleep hours?	Scatter plot — Daily Screen Time vs Sleep Hours
6	How does average gaming time compare with social-media usage?	Clustered column — Gaming vs Social Media
7	Does weekend screen time differ between addicted and non-addicted users?	Bar chart — Avg Weekend Screen Time by Addicted Status
8	How can the report be filtered dynamically by gender, age group, addiction level and stress?	Slicers on all analytical pages

4.2 Developing Visualizations

The visualization page brings the core charts together on a single exploratory canvas with gender, addiction-level and age-group slicers, so each question above can be answered interactively.

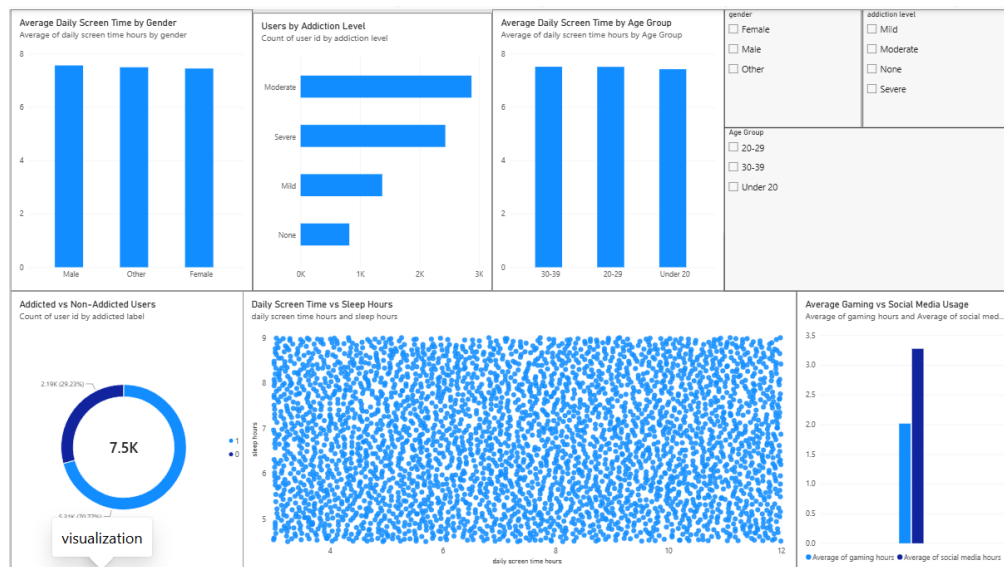


Figure 1 — Visualization page: gender, addiction level, age group, donut, scatter and gaming vs social media

- Average daily screen time is almost identical across Male (7.6), Other (7.5) and Female (7.4) — gender is not a differentiator.
- The donut confirms the headline split: 5.31K addicted (70.77%) against 2.19K non-addicted (29.23%) out of 7.5K users.
- The scatter of daily screen time against sleep hours forms a dense uniform cloud, showing no simple linear relationship at the individual level — the effect appears only when averaged across groups.
- Average social-media usage (3.27 hrs) is over 1.6× average gaming time (2.01 hrs), making social media the larger single contributor to daily screen time.

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

The Digital Well-Being & Addiction Insights dashboard applies a dark indigo/cobalt theme, places four high-level KPI cards along the top margin, and anchors an addiction-level drop-down slicer on the control bar so every visual below cross-filters together.

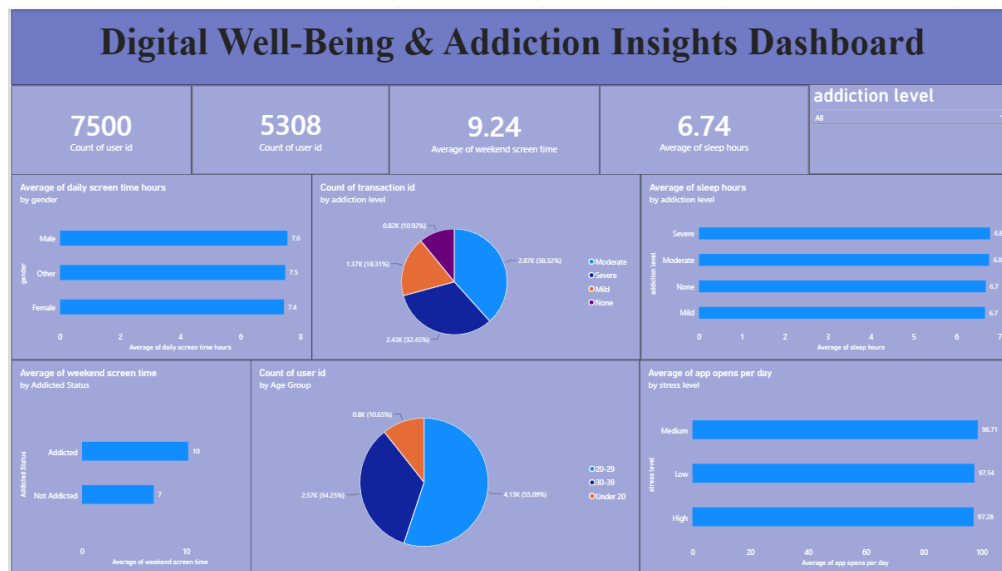


Figure 2 — Dashboard: Digital Well-Being & Addiction Insights

- Scale: 7,500 users tracked in total, of whom 5,308 are addicted; mean weekend screen time is 9.24 hrs and mean sleep is 6.74 hrs.
- Addiction profile: Moderate is the largest share at 38.32%, Mild 32.45%, Severe 18.31% and None 10.92% — roughly nine in ten users sit at Mild or above.
- Weekend gap: addicted users average 10 hrs of weekend screen time against 7 hrs for non-addicted users — a ~3-hour gap, the sharpest behavioural separator on the dashboard.
- Demographics: the 20–29 cohort dominates at 55.09%, followed by 30–39 at 34.25% and Under 20 at 10.65%, so interventions should target young working adults.
- Sleep is flat across addiction levels (6.7–6.8 hrs) and app opens are flat across stress levels (97.28–98.71), indicating that volume of interaction alone does not explain stress or sleep outcomes.

6. Report

6.1 Report Design Screenshot with Insights

Two analytical report pages extend the dashboard. Report 1 focuses on interaction volume and the weekend-usage escalation, and includes a decomposition tree for guided drill-down. Report 2 profiles stress and sleep and uses the Key Influencers AI visual to test what drives the Addicted status.

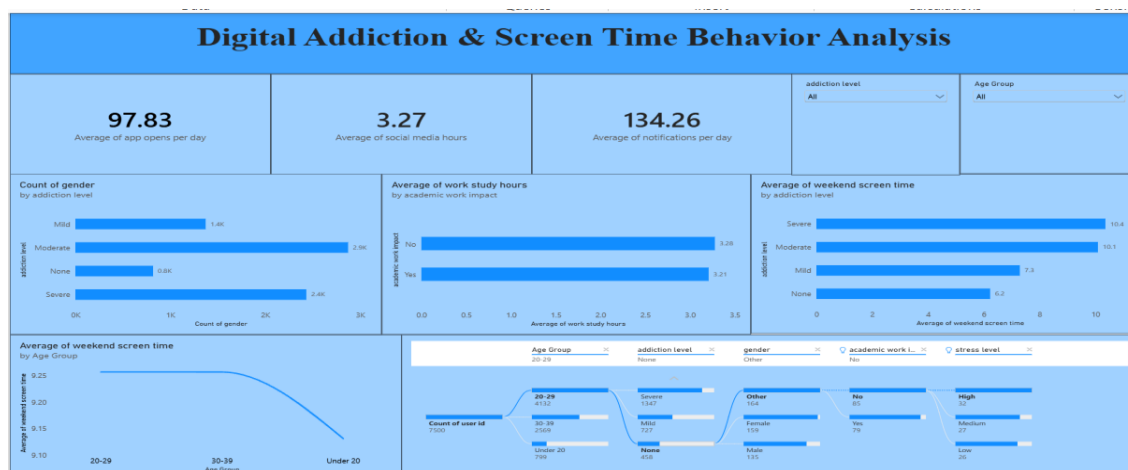


Figure 3 — Report 1: Digital Addiction & Screen Time Behaviour Analysis

- Interaction volume is high: 97.83 app opens and 134.26 notifications per user per day, alongside 3.27 hrs of social media.
- Weekend screen time escalates cleanly with addiction level — None 6.2, Mild 7.3, Moderate 10.1 and Severe 10.4 hrs — confirming weekend behaviour as the clearest marker of severity.
- Average work/study hours barely differ between users reporting academic/work impact (3.21) and those who do not (3.28), so self-reported impact is not explained by workload.
- The decomposition tree lets the analyst unpack the 7,500 users down through Age Group → addiction level → gender → academic impact → stress level; the 20–29 branch alone holds 4,132 users.

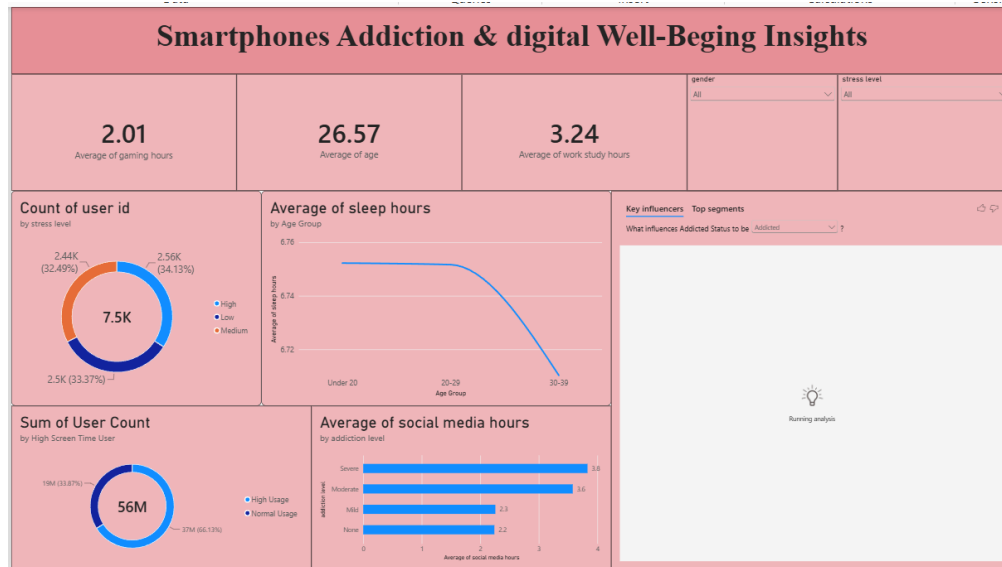


Figure 4 — Report 2: Smartphone Addiction & Digital Well-Being Insights

- Stress is split almost evenly — High 34.13%, Medium 32.49% and Low 33.37% — so stress is broadly distributed rather than concentrated.
- Social-media hours rise consistently with addiction level: None 2.2, Mild 2.3, Moderate 3.6 and Severe 3.8 hrs, making social media the strongest single app-category signal.
- Average sleep declines slightly from Under 20 through to the 30–39 group, a small but consistent downward trend.
- The High Screen Time User split shows roughly one third of the weighted user base in the High Usage band against two thirds in Normal Usage.

7. Performance Testing

7.1 Utilization of Data Filters

Eight slicer visuals are distributed across the four analytical pages, supported by page-level and visual-level filters in the Filters pane.

Filter / Slicer	Type	Pages Applied	Purpose
gender	List / drop-down slicer	Visualization, Report 2	Compare behaviour across Male, Female and Other
addiction level	List / drop-down slicer	Visualization, Dashboard, Report 1	Isolate None, Mild, Moderate and Severe cohorts
Age Group	List / drop-down slicer	Visualization, Report 1	Focus on Under 20, 20–29 or 30–39
stress level	Drop-down slicer	Report 2	View results for Low, Medium or High stress
Cross-highlighting	Interaction	All pages	Selecting a chart segment filters the rest of the canvas

Filter / Slicer	Type	Pages Applied	Purpose
Decomposition tree	Drill-down	Report 1	Path-based drill through up to five attributes

7.2 No. of Calculation Columns / Measures

Seven calculated columns were created, alongside the implicit aggregations (Count, Average, Sum) used by the cards and charts.

#	Calculated Column	Purpose
1	Age Group	Buckets age into Under 20, 20–29 and 30–39
2	Addicted Status	Converts the addicted_label code into "Addicted" / "Not Addicted"
3	High Screen Time User	Flags each row as "High Usage" or "Normal Usage"
4	User Count	Fixed counter used for total user cards and donuts
5	Academic/Work Impact Level	Tags impact as High, Medium or Low Impact
6	Stress Category	Groups stress into Low, Medium and High Stress
7	Sleep Category	Labels Low Sleep (under 6 hrs) or Normal Sleep (6+ hrs)

7.3 No. of Visualizations

The .pbix contains 55 visual containers across 5 pages — 43 data visuals, 8 slicers and 4 title textboxes.

Page	Visuals	Composition
KPI	11	11 card visuals
Visualization	9	3 clustered column, 1 clustered bar, 1 donut, 1 scatter, 3 slicers
Dashboard	13	4 cards, 4 clustered bar, 2 pie, 1 slicer, 2 textboxes
Report 1	11	3 cards, 3 clustered bar, 1 line, 1 decomposition tree, 2 slicers, 1 textbox
Report 2	11	3 cards, 2 donut, 1 clustered bar, 1 line, 1 key influencers, 2 slicers, 1 textbox
Total	55	43 data visuals + 8 slicers + 4 textboxes

The KPI page consolidates the eleven headline metrics that every other page is built on.

5308 Count of user id	7500 Count of user id	26.57 Average of age
7.50 Average of daily screen time hours	97.83 Average of app opens per day	2.01 Average of gaming hours
6.74 Average of sleep hours	134.26 Average of notifications per day	3.27 Average of social media hours
9.24 Average of weekend screen time	3.24 Average of work study hours	

Figure 5 — KPI summary page: 11 card visuals

8. Conclusion / Observation

The analysis answers both problem statements. On PS-1, the scale of the issue is confirmed: 5,308 of 7,500 users (70.77%) are classified as addicted, and 89% of users sit at Mild severity or above, with Moderate the single largest

band at 38.32%. The affected population is concentrated in the 20–29 age group, which alone accounts for 55.09% of users.

On PS-2, the factors separate clearly. Weekend screen time is the strongest discriminator — addicted users average 10 hrs against 7 hrs for non-addicted users, climbing monotonically from 6.2 hrs at None to 10.4 hrs at Severe. Social-media hours follow the same pattern (2.2 hrs at None to 3.8 hrs at Severe), while gaming stays far lower at 2.01 hrs. Notification and app-open volume is high overall (134.26 and 97.83 per day) but almost identical across stress levels, acting as constant pressure rather than a differentiator.

Conversely, several intuitive factors proved weak: gender differences in daily screen time are negligible (7.4–7.6 hrs), average sleep barely moves across addiction levels (6.7–6.8 hrs), and the scatter shows no individual-level link between screen time and sleep. The practical conclusion is that well-being efforts should target weekend usage and social-media sessions among 20–29 year olds rather than notification counts or a particular gender.

9. Future Scope

- Extend the dataset with time-stamped session logs so that trends can be analysed by day, week and time of day instead of aggregated averages.
- Add a predictive layer — a classification model scoring each user's probability of becoming addicted — and surface the score as a Power BI measure.
- Publish the report to the Power BI Service with scheduled refresh, and enable the mobile layout for on-the-go access.

10. Appendix

10.1 Source Code

SQL used against the MySQL staging copy for exploratory analysis and data-quality validation.

```
USE smartphone;
SELECT * FROM smartphone_usage;
SELECT gender, COUNT(*),
       SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage
GROUP BY gender;
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours),
       AVG(stress_level = 'High')
FROM smartphone_usage;
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL
      OR addiction_level IS NULL;
SELECT addicted_label, stress_level, COUNT(*)
FROM smartphone_usage
GROUP BY addicted_label, stress_level
ORDER BY addicted_label, stress_level;
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours),
       AVG(sleep_hours)
FROM smartphone_usage
GROUP BY addiction_level;
SELECT academic_work_impact, AVG(daily_screen_time_hours),
       AVG(social_media_hours), AVG(gaming_hours)
FROM smartphone_usage
GROUP BY academic_work_impact;
```

10.2 GitHub & Project Demo Link

Item	Link / Reference
GitHub Repository	https://github.com/ChokkaraMeghana/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics.git
Power BI File	smartphone.pbix
SQL Source File	smartphone.sql
Project Demo Video	https://drive.google.com/file/d/1e2DeJ-E7nU9tBme-U_pWdyF3b92RbSqr/view?usp=sharing